Domain Understanding – F1 Race Winner Prediction

Conducted by: Tihomira Miteva

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1. Central Research Question

How can historical Formula 1 race data, including track conditions, qualifying performance, driver form, team performance, and strategic decisions, be used to predict the three drivers with the highest probability of winning a Grand Prix?

This question supports identifying the most relevant performance and environmental variables while ensuring the model produces ranked, probabilistic race outcome predictions rather than single deterministic results.

2. Exploratory Research

To build understanding of the domain, exploratory research was conducted using:

- historical Formula 1 race data and known F1 strategy principles,
- analysis of factors influencing race outcomes (track type, weather, qualifying position, tire strategy),
- trends in driver and constructor performance across seasons,
- publicly available datasets suitable for machine learning.

The main data source used is the **F1DB open-source repository** (https://github.com/f1db/f1db), which provides structured datasets for races, drivers, constructors, qualifying results, pit stops, and standings. This repository is widely used for motorsport analytics and offers consistent historical data.

Exploration of both the domain and the data shows that race outcomes are influenced by a **combination of driver performance, team strength, grid position, strategic decisions, and environmental conditions**, rather than by a single dominant factor.

3. Analytic Approach

3.1 Target Variable

The target variable is defined as **Winning Probability (0–1)** for each driver in a given race.
This continuous value represents the likelihood that a driver wins a specific Grand Prix based on available performance, team, and track data.

This allows drivers to be ranked rather than classified into a single winner, which better reflects the uncertainty and competitive nature of Formula 1 racing.

3.2 Type of Problem

This is primarily a **multi-class probabilistic prediction problem**, where each driver in a race is assigned a probability score.

Alternative approaches include:
• **Classification:** predicting a single race winner.
• **Ranking:** ordering all drivers by predicted chance of winning.

For this project, probabilistic ranking is preferred because it supports selecting the top three most likely winners rather than producing a single deterministic result.

Potential Machine Learning Models

Linear Regression
• Why use it: Serves as a baseline model to test whether race outcomes show simple linear relationships with variables such as grid position or championship points.
• Good for: Understanding feature influence before applying more complex models.

Decision Trees
• Why use it: Captures rule-based relationships such as "If grid position is high and team ranking is top-tier, then winning probability increases."
• Good for: Interpretability and understanding how strategic and performance factors interact.

Random Forest Regressor
• Why use it: Combines multiple decision trees to improve stability and prediction accuracy.
• Good for: Feature importance analysis and reducing overfitting to individual race scenarios.

K-Nearest Neighbors (KNN)
• Why use it: Finds races similar in track type, grid positions, and team performance to estimate likely outcomes.
• Good for: Pattern-based prediction using historical race similarity.

The final model selection will prioritize prediction accuracy and explainability.

3.4 Evaluation Metrics

• **Mean Absolute Error (MAE):** Measures how close predicted probabilities are to actual race outcomes.
• **R^2 Score:** Evaluates how well performance variables explain variance in winning probability.
• **Top-K Accuracy:** Checks whether the actual race winner appears in the model's top three predicted drivers.

4. Data Requirements

4.1 Objectives

The project aims to:
• Predict the most likely race winners using real performance and strategy data.
• Support explainable, probability-based predictions rather than fixed outcomes.
• Identify which performance factors have the strongest impact on winning chances.

Success is measured by ranking accuracy and the model's ability to consistently place real winners in the top predicted group.

4.2 Characteristics

The required data includes:
• Track data: location, circuit type, layout characteristics, and weather indicators.
• Driver data: championship points, season form, qualifying position, historical track performance.
• Team data: constructor standings, car reliability, pit stop performance.
• Strategy data: tire compounds, number of pit stops, and pit timing.

All variables will be numerical or encoded into numerical formats for model compatibility.

4.3 Sources

• F1DB open-source repository
• Public motorsport analytics datasets
• Historical Formula 1 race records

4.4 Legality & Ethics

• All data is publicly available and non-personal.
• Predictions are presented as probabilities, not guarantees.
• The model is used for analysis and learning purposes and does not influence real-world competition.

4.5 Version Control

Version control is maintained through a Git repository to track dataset changes, feature engineering, and model development. This ensures transparency and reproducibility throughout the project.

5. Iterative Process

• **Proposal Phase:** Define research question and identify data sources.
• **Provisioning Phase:** Collect, clean, and prepare race, driver, and constructor data.
• **Model Development:** Train and test multiple models and tune performance.
• **Evaluation & Delivery:** Validate ranking accuracy, document findings, and finalize predictions.

6. Conclusion

This document outlines the approach for predicting Formula 1 race winners using historical performance, strategic data, and track conditions. By framing the problem as a probabilistic ranking task, the model reflects the competitive and uncertain nature of motorsport.

The foundation established here ensures that predictions are explainable, data-driven, and aligned with real racing dynamics, supporting meaningful analysis of how drivers, teams, and strategies influence race outcomes.